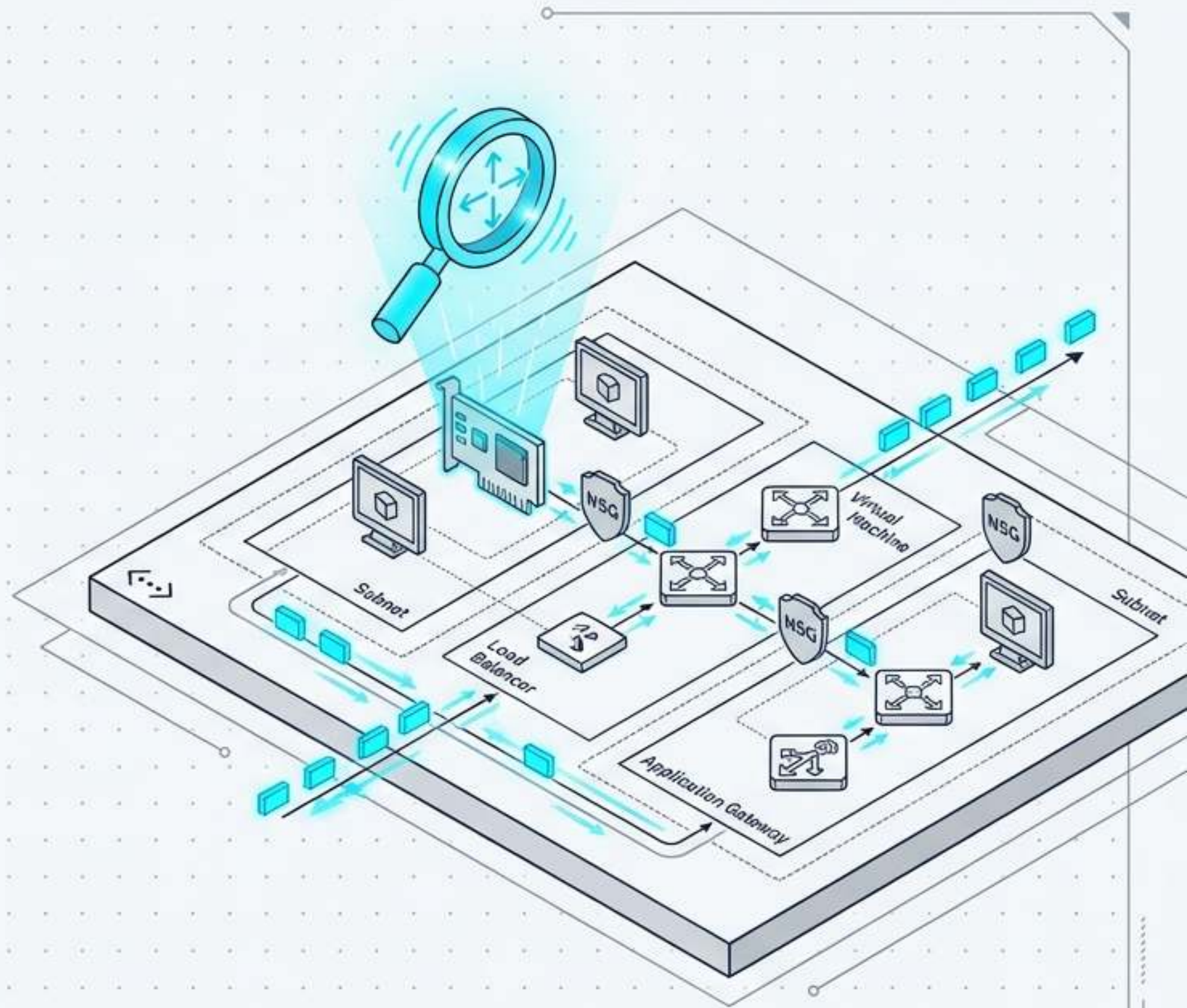


AZ-500 Exam Prep Series

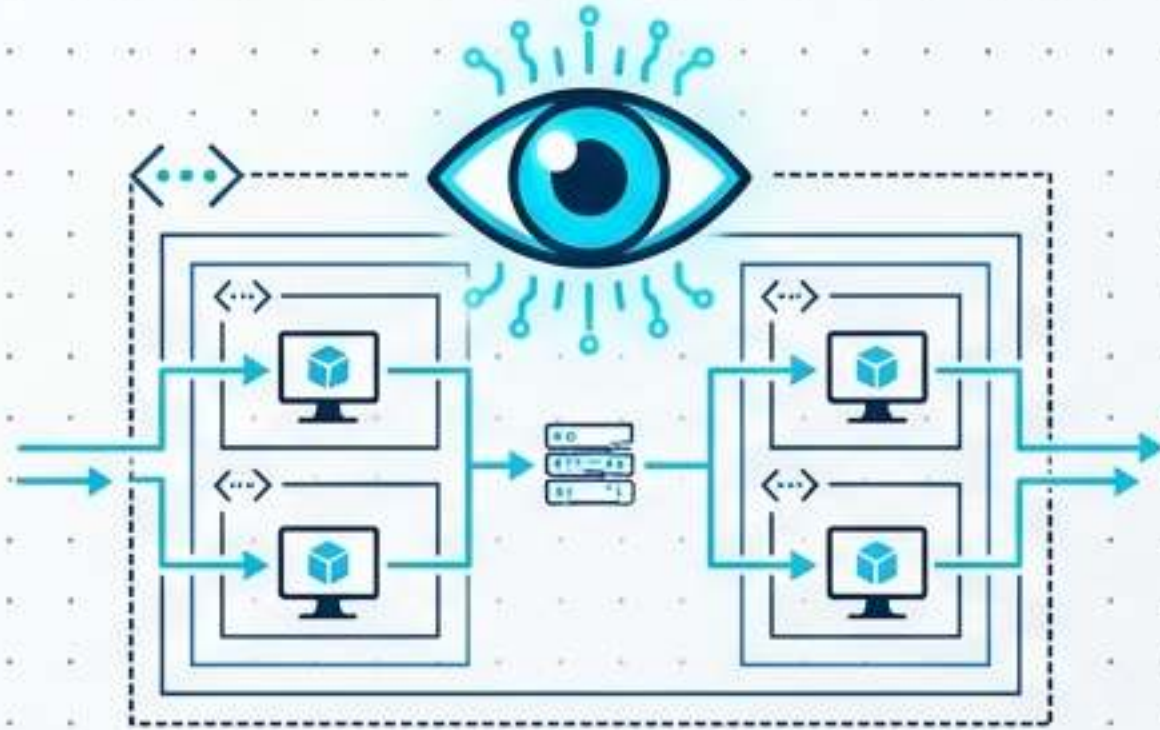
Network Watcher & Diagnostics

The AZ-500 Master Reference Guide



The Diagnostic Observer, Not the Enforcer

The Observer



Network Watcher. A regional service used to monitor, diagnose, and gain insights into network health within a VNet. It observes and proves what is happening but cannot change security rules itself.

The Enforcers

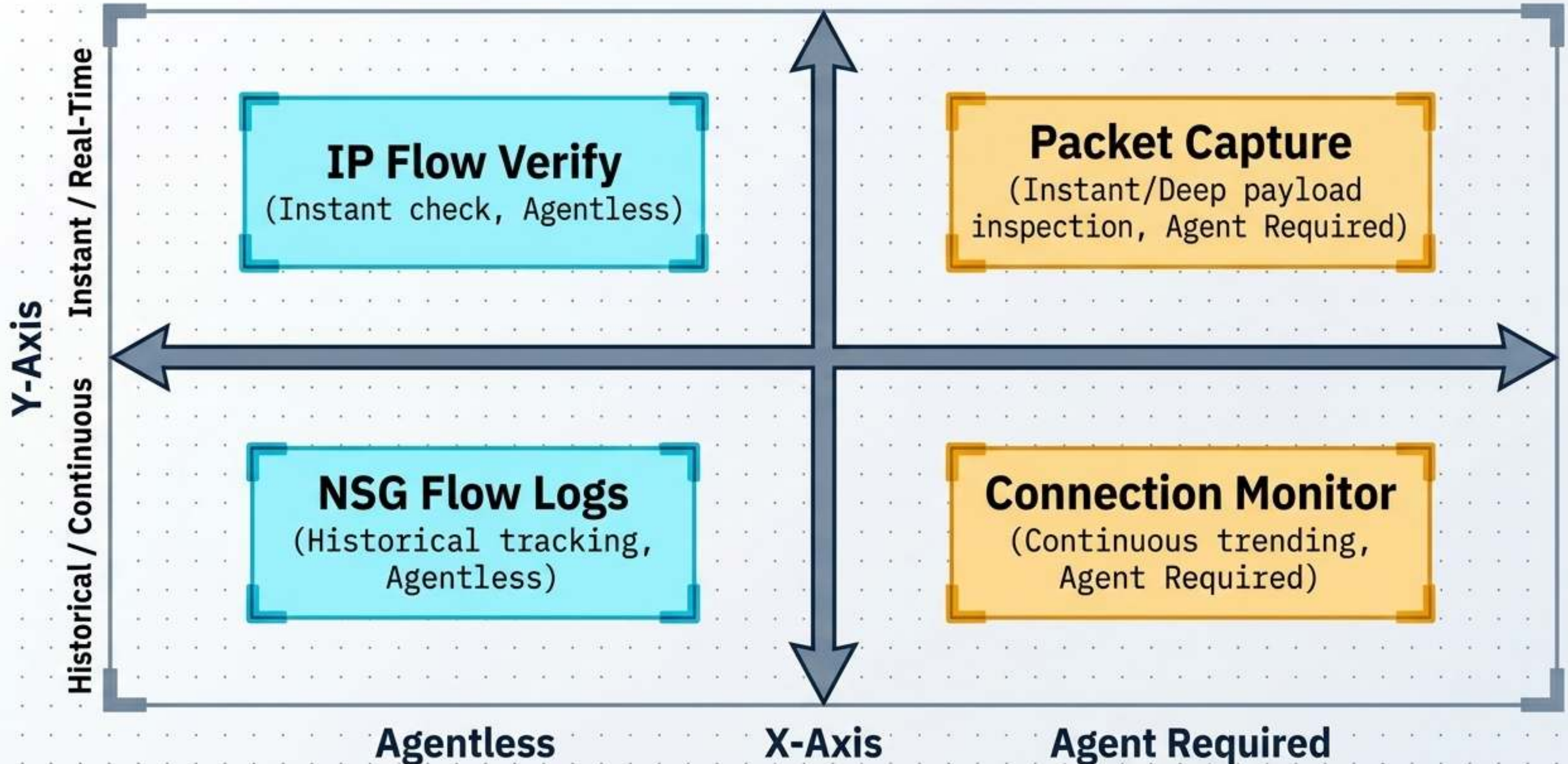


NSGs, ASGs, & AVNM. These actual enforcement mechanisms drop or allow packets. **Network Watcher** only reports on their behavior.



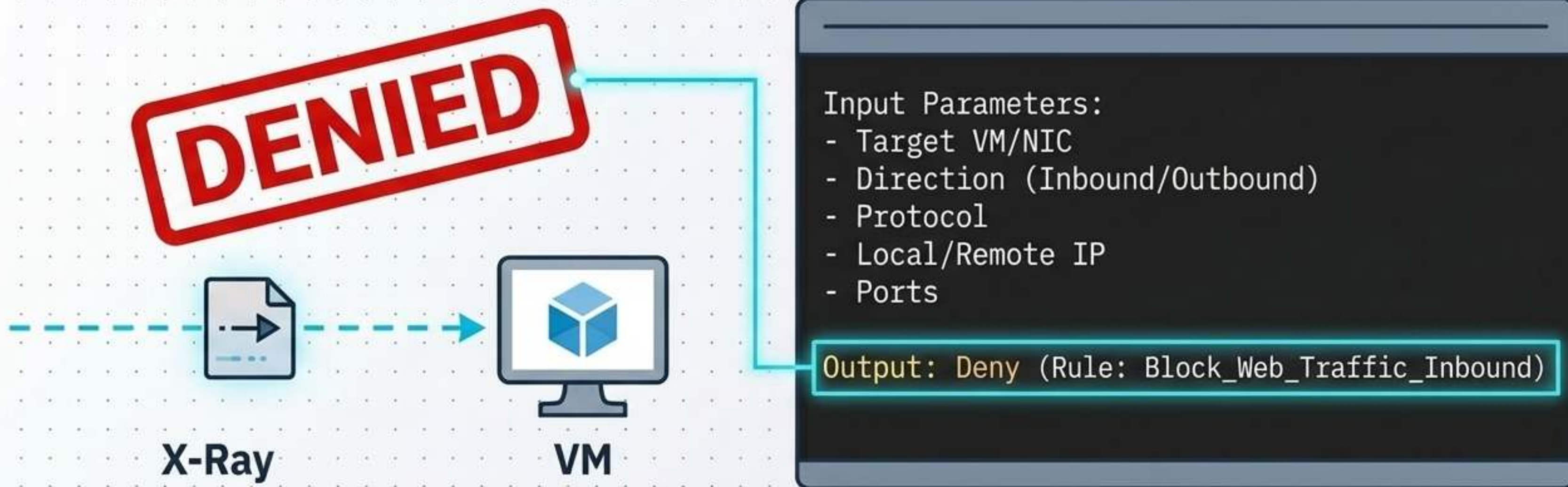
CRITICAL TRAP: Do not select Network Watcher if a scenario asks how to block or prevent traffic.

The Diagnostic Quadrant



IP Flow Verify: The Instant Rule Check

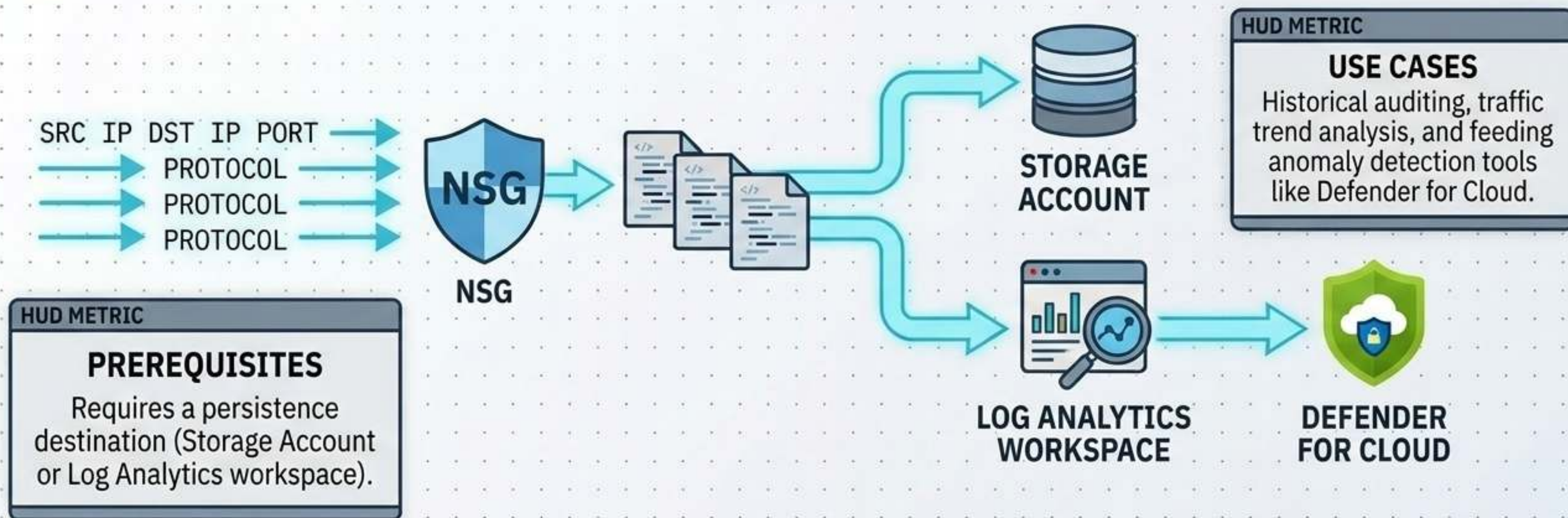
Would this specific packet be allowed or denied RIGHT NOW, and by which rule?



NO VM AGENT EXTENSION REQUIRED.

NSG Flow Logs: The Historical Traffic Audit

What traffic has actually passed through this NSG historically?



TRACKS 5-TUPLE FLOW DATA, NOT PACKET PAYLOAD.

Connection Monitor: Continuous Health Tracking

HUD BANNER

Is connectivity between point A and B healthy over time?



HUD METRIC CARD

Purpose



Continuous, ongoing monitoring of latency and reachability trends.

HUD METRIC CARD

Use Cases



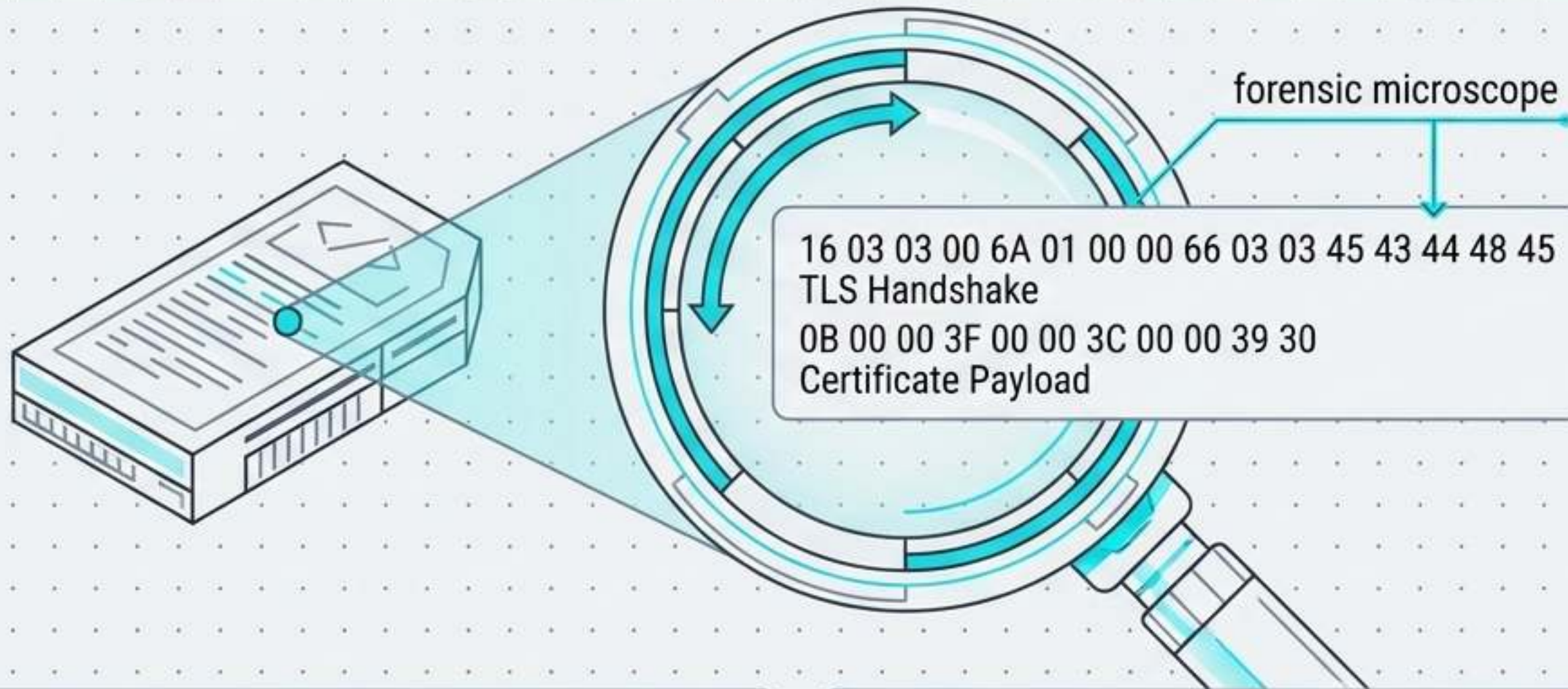
Branch-to-Azure stability, hybrid connectivity health.



REQUIRES NETWORK WATCHER AGENT EXTENSION

Packet Capture: Deep Payload Inspection

What is literally inside the packets on this NIC?



Purpose

Deep troubleshooting for complex, application-layer issues (TCP handshake failures, TLS certificate exchange problems).

Output

```
> Generating .cap or .pcap file...  
File saved for analysis in external tools like Wireshark.
```

⚠ REQUIRES AGENT EXTENSION (Triggered manually or by alerts).

AZ-500 Tool Selection Matrix

Tool	Question It Answers	Timeline	Agent Req?	Exam Keyword Phrase
IP Flow Verify	Is it blocked right now?	Real-time	No	Verify why a path is denied
NSG Flow Logs	Who connected last week?	Historical	No	Analyze traffic over the last week
Connection Monitor	Is the hybrid link stable?	Continuous	Yes	Monitor ongoing latency trends
Packet Capture	Why is the TLS handshake failing?	Real-time/Deep	Yes	Inspect raw traffic / debug headers

Scenario Playbook: Instant vs. Historical

Ticket 1 (The Blocked Dev)



A user cannot reach a SQL database instantly. You need to know exactly which rule is dropping the traffic.



IP Flow Verify

Instantly checks current rules without needing an agent on the VM.

Ticket 2 (The Audit Request)



An auditor asks for a list of all IP addresses that connected to a specific secure subnet over the entire last month.



NSG Flow Logs

Queries historical connection data persisted in a Storage Account.

Scenario Playbook: Continuous vs. Payload

Ticket 3 (The Shaky Branch)



You need to track if a branch-to-Azure VPN connection remains stable and trigger an alert if latency spikes above 50ms.



Connection Monitor

Provides continuous, trend-based health tracking over time.

Ticket 4 (The Broken Cert)



You need to inspect raw traffic to debug a failing certificate exchange between a web app and a backend API.



Packet Capture

The only tool that looks inside the packet at the TLS payload.

Critical AZ-500 Traps



Trap 1: Timeline Confusion

Do not confuse **IP Flow Verify** (**instant/packet-level rule check**) with **NSG Flow Logs** (**historical/long-term traffic trends**). If the scenario says "last week" or "history," it is never IP Flow Verify.



Trap 2: Action Confusion

Do not select **Network Watcher** to **enforce** security.

If the scenario asks how to "**prevent**" or "**block**" traffic, look for **NSG, ASG, AVNM, or Azure Firewall**. Network Watcher only reports.

The Diagnostic Decision Tree

